

ServiceNow Project Documentation

Project: Streamlining Ticket Assignment For Efficient Support Operations

Team Details

Team ID	LTVIP2026TMIDS80714
Team Leader	Venu Gopala Swamy Manikala
Team Member 1	Bendukuri Tapaswi Moses
Team Member 2	Ganisetty Eswar
Team Member 3	Durga Naga Saisri

Project Overview

This project implements a safeguard in ServiceNow to prevent deletion of a user record when the user is assigned to one or more Incident records. The solution is implemented using a Business Rule on the **User [sys_user]** table, triggered **Before Delete**. If an assignment exists, the delete operation is aborted and an error message is displayed.

Objectives

- Prevent accidental deletion of users who are still working on active incidents.
- Maintain ITSM process continuity and assignment integrity.
- Provide clear feedback to administrators via an error message.

1. Configuration & Setup

The project uses a server-side Business Rule in the Global application scope. The rule is created on the User table and configured to run on delete.

1.1 Business Rule: When to run

Table: User [sys_user]

When: Before

Delete: True

Order: 100

The screenshot shows the ServiceNow interface for configuring a Business Rule named 'Prevent User Deletion'. The rule is set to run on the 'User [sys_user]' table. The 'When' tab is selected, showing the rule should run 'before' the action with an 'Order' of 100. The 'Delete' checkbox is checked under the 'When to run' section. The 'Advanced' tab is also visible, showing the script area.

Business Rule - Prevent User Deletion

Name: Prevent User Deletion
Table: User [sys_user]
Application: Global
Active: ☒
Advanced: ☒

When to run | Actions | Advanced

Specify whether the business rule should run on **Insert** or **Update**. Use **Filter Conditions** to specify under which conditions the business rule should run.

When: before
Order: 100

Insert: ☐
Update: ☐
Delete: ☒
Query: ☐

Filter Conditions: [Add Filter Condition](#) [Add OR Clause](#)

1.2 Business Rule: Script (Advanced tab)

The Business Rule checks the Incident table for any record where **Assigned to** equals the user being deleted. If at least one record exists, the script shows an error message and aborts the delete action.

The screenshot shows the 'Advanced' tab of the Business Rule configuration. The 'Script' section is active, displaying a JavaScript function that checks for incidents assigned to the user being deleted. The script uses the GlideRecord class to query the Incident table and aborts the action if any records are found.

Business Rule - Prevent User Deletion

When to run | Actions | **Advanced**

Condition:

Script: ☒ Turn on ECMAScript 2021 (ES12) mode

```
1 (function executeRule(current, previous /*null when async*/) {  
2  
3   var incGr = new GlideRecord('incident');  
4   incGr.addQuery('assigned_to', current.sys_id);  
5   incGr.setLimit(1);  
6   incGr.query();  
7  
8   if (incGr.next()) {  
9     gs.addErrorMessage('This user cannot be deleted because they are assigned to one or  
10    more incidents.');
```

11 current.setAbortAction(true);
12 }
13 }}(current, previous);

[Update](#) [Delete](#)

Related Links
[Run Point Scan](#)

2. Test Data Creation

Two users were created to validate the functionality: one user that is assigned to an incident (deletion must be blocked), and one unused user (deletion should succeed).

2.1 User 1 Created

The screenshot shows the ServiceNow user profile page for 'User - Rudraraju Jogi Subrahmanya...'. The 'User' tab is selected, displaying the following fields and values:

- User ID: raju
- First name: Rudraraju
- Last name: Jogi Subrahmanya Raju
- Title: (empty)
- Department: (empty)
- Email: rudrarajogisubrahmanyamraju@gn
- Language: -- None --
- Calendar integration: Outlook
- Time zone: System (America/Los_Angeles)
- Date format: System (yyyy-MM-dd)
- Business phone: (empty)
- Mobile phone: (empty)
- Photo: Click to add...
- Password needs reset: ☐
- Locked out: ☐
- Active: ☒
- Identity type: Human
- Internal Integration User: ☐

Buttons at the bottom include 'Update', 'Set Password', and 'Delete'. Below the form, there are 'Related Links' (View linked accounts, View Subscriptions, Reset a password) and a tabbed interface with 'Roles (43)' selected.

2.2 Roles Assigned to User 1

Roles are visible in the Roles related list on the user record. (Roles are not required for the Business Rule logic, but are part of standard user setup.)

The screenshot shows the ServiceNow user profile page for 'User - Rudraraju Jogi Subrahmanya...' with the 'Roles' tab selected. The table below lists the roles assigned to the user.

Role	State	Inherited	Inheritance Count
sn_request_read	Active	true	1
sn_itam_recomm.recommendations_read	Active	true	1
sn_sow.it_agent_dashboard_user	Active	true	1
sn_incident_read	Active	true	1
sn_gd_guidance.guidance_user	Active	true	7
viz_creator	Active	true	1
data_manager_user	Active	true	1
interaction_agent	Active	true	6
cmdb_ms_user	Active	true	1
sn_request_approver_read	Active	true	1
certification	Active	true	1
sn_sow.sow_list	Active	true	5
sn_change_read	Active	true	1

3. Incident Assignment Validation

An incident was created and assigned to User 1. The incident was set to **In Progress** to represent an active assignment scenario.

3.1 Incident Assigned to User 1

Expected fields shown:

- Assigned to = User 1
- State = In Progress

The screenshot displays the ServiceNow interface for an incident record. The top navigation bar includes the ServiceNow logo and tabs for All, Favorites, History, and Admin. The incident title is "Incident - INC0010003". Below the title, there are action buttons: Discuss, Follow, Update, Resolve, and Delete. The form is divided into two main sections. The left section contains fields for Number (INC0010003), Caller (Rudraraju Jogi Subrahmanyam Raju), Category (Inquiry / Help), Subcategory (-- None --), Service, Service offering, Configuration item, and Short description (Test incident for user deletion validation). The right section contains fields for Channel (-- None --), State (In Progress), Impact (3 - Low), Urgency (3 - Low), Priority (5 - Planning), Assignment group, and Assigned to (Rudraraju Jogi Subrahmanyam Raju). At the bottom, there are tabs for Notes, Related Records, and Resolution Information, along with Watch list and Work notes list sections.

servicenow All Favorites History Admin Incident - INC0010003 Search

Incident INC0010003 Discuss Follow Update Resolve Delete

Number INC0010003 Channel -- None --

* Caller Rudraraju Jogi Subrahmanyam Raju State In Progress

Category Inquiry / Help Impact 3 - Low

Subcategory -- None -- Urgency 3 - Low

Service Priority 5 - Planning

Service offering Assignment group

Configuration item Assigned to Rudraraju Jogi Subrahmanyam Raju

* Short description Test incident for user deletion validation

Description

Related Search Results

Notes Related Records Resolution Information

Watch list Work notes list

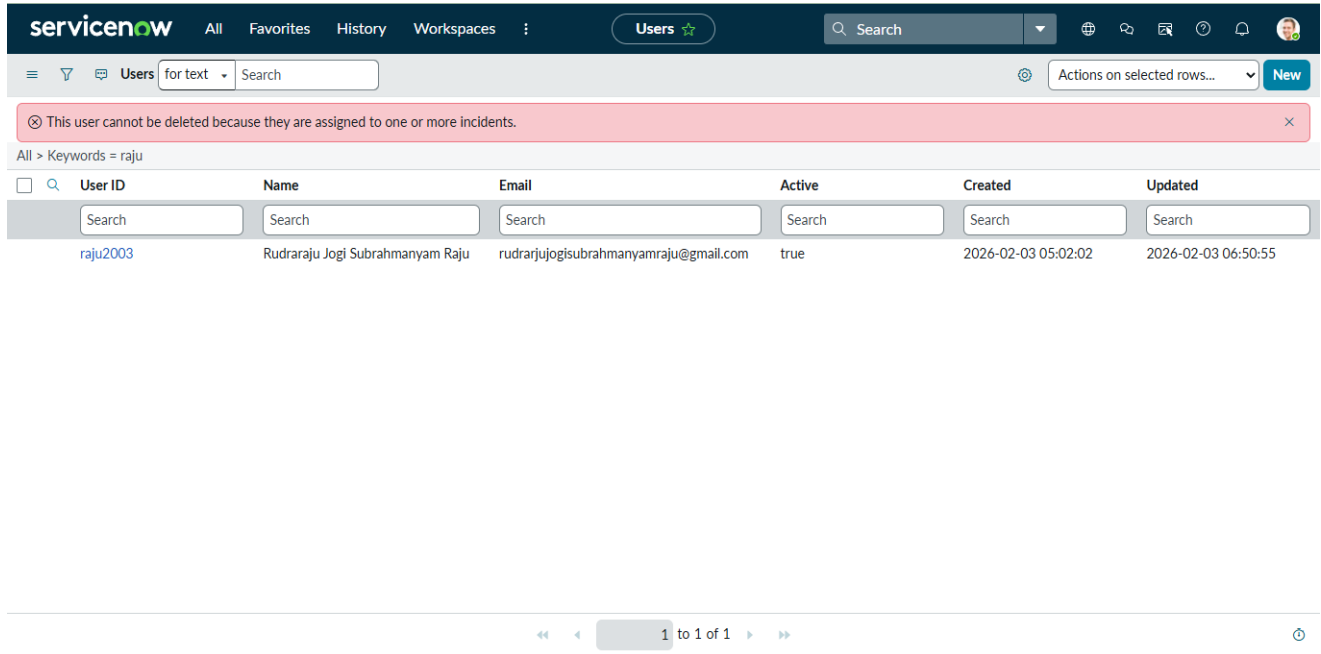
Work notes Work notes

4. Testing & Results

The deletion behavior was tested for both users to confirm correct Business Rule functionality.

4.1 Attempt to Delete Assigned User (User 1)

When trying to delete User 1 (who is assigned to an incident), ServiceNow blocks the deletion and shows an error message.

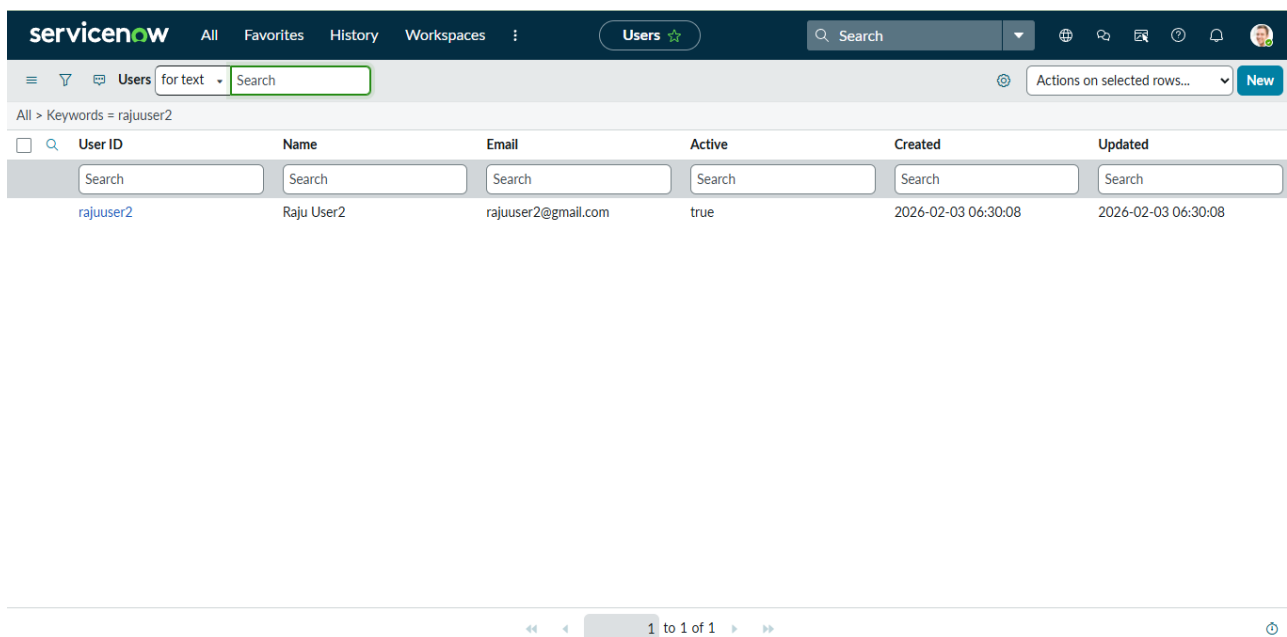


The screenshot shows the ServiceNow 'Users' list interface. A red error banner at the top states: 'This user cannot be deleted because they are assigned to one or more incidents.' Below the banner, the table shows one user: 'raju2003' (Rudraraju Jogi Subrahmanyam Raju). The pagination at the bottom indicates '1 to 1 of 1'.

User ID	Name	Email	Active	Created	Updated
raju2003	Rudraraju Jogi Subrahmanyam Raju	rudrarajuogisubrahmanyamraju@gmail.com	true	2026-02-03 05:02:02	2026-02-03 06:50:55

4.2 Delete Unused User (User 2)

When trying to delete User 2 (not assigned to any incident), the deletion succeeds normally. This confirms that the Business Rule only blocks deletion when assignment exists.



The screenshot shows the ServiceNow 'Users' list interface. The search filter is set to 'rajuuser2'. The table shows one user: 'rajuuser2' (Raju User2). The pagination at the bottom indicates '1 to 1 of 1'.

User ID	Name	Email	Active	Created	Updated
rajuuser2	Raju User2	rajuuser2@gmail.com	true	2026-02-03 06:30:08	2026-02-03 06:30:08

5. Business Rule Logic (Explanation)

The Business Rule executes on the User table before delete. It performs a query on the Incident table using the sys_id of the user being deleted.

```
var incGr = new GlideRecord('incident');
incGr.addQuery('assigned_to', current.sys_id);
incGr.setLimit(1);
incGr.query();

if (incGr.next()) {
    gs.addErrorMessage('This user cannot be deleted because they are assigned to one or more incidents.');
```

current.setAbortAction(true);

```
}
```

Why this works

- The query checks if at least one incident exists for the user.
- If found, deletion is blocked using **current.setAbortAction(true)**.
- The administrator receives a clear error note using **gs.addErrorMessage()**.

6. Conclusion

This project successfully prevents deletion of users who are still assigned to incidents. It improves data integrity, reduces operational risk, and ensures incident ownership is preserved in ServiceNow.